

# Model Memory vs Retrieval

Project: Verified Study Buddy

## 1. Your mission

### Your learning target

I can explain the mechanism and terminology of retrieval and indexing.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

### Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Verify Numbers and Dates. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Hallucination Failure Types. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Visual Composition. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Week 17: Model Memory vs Retrieval

Learned parameters are not a searchable source library.



### CREATOR MISSION

Decide when to answer, retrieve, calculate, or abstain.

Creator name and date:

**Write or draw your reasoning and evidence.**

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### Readiness for the connected explanation

Recall Speech pipelines and multimodal evidence. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

### This week's end goal

By the end, I can explain model memory vs retrieval using a traced example and a stated limitation.

## 2. Notice the evidence

### PICTURE STORY

Notice → Predict → Explain

Begin with visible evidence. Do not explain more than the picture and information support.



MYSTERY

The learners must use model memory vs retrieval to solve a problem. What information do they have? What decision or calculation do they make? What would count as evidence that it worked?

Three observations (what you can point to):

Write or draw your reasoning and evidence.

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My prediction and the clue supporting it:

Write or draw your reasoning and evidence.

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## A closer explanation

Retrieval returns stored records selected for a query. An index is a searchable representation of a versioned corpus; generated wording is a later operation.

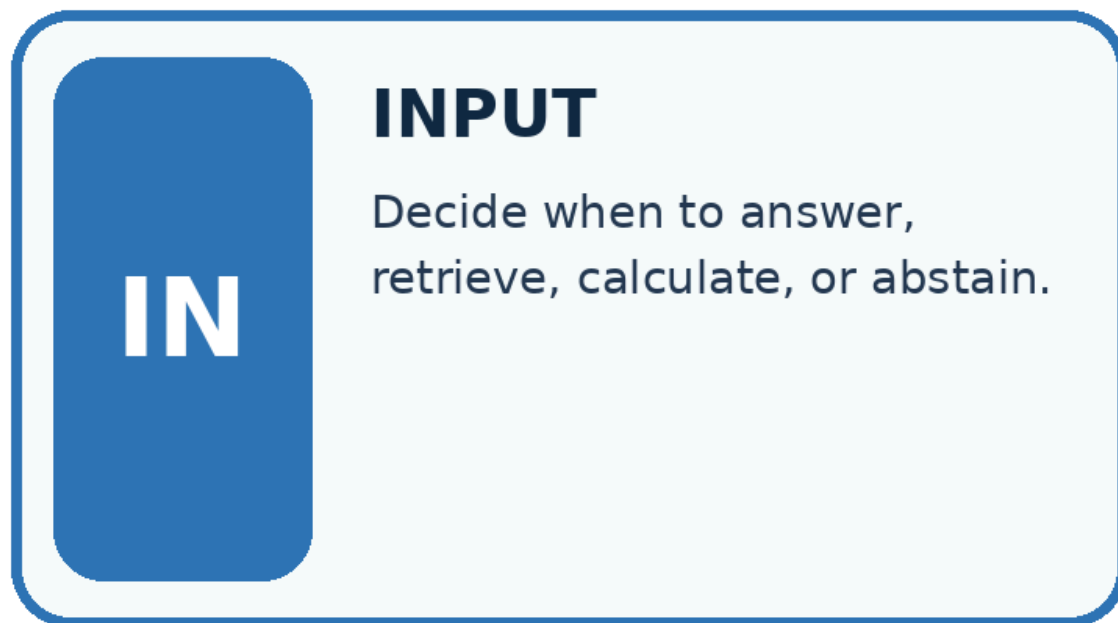
Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing. These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

## 3. See how it works

VISUAL MECHANISM

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.



↓



## SYSTEM

Learned parameters are not a searchable source library.



OUT

## EVIDENCE

Decide when to answer, retrieve, calculate, or abstain.

Which stage could fail even when the other two are correct? Explain:

**Write or draw your reasoning and evidence.**

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## A closer explanation

Assign stable document IDs, versions and permitted access before indexing. Retrieve only authorised records. Preserve source references and define behaviour when a document is missing, outdated or access-restricted.

## Retrieval and indexing — evidence trace

### Input and scope · Check or calculation · Conclusion and limit

**Input and scope:** Index v1 contains notice N1 for room A. Corpus v2 supersedes it with N2 for room B, but the query still returns N1. What can be reported?

**Check or calculation:** Assign stable document IDs, versions and permitted access before indexing. Retrieve only authorised records. Preserve source references and define behaviour when a document is missing, outdated or access-restricted.

**Conclusion and limit:** The stale index returns old evidence. It cannot establish current room approval; refresh against authorised corpus v2 or report unresolved freshness.

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The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same inputs. If a required input is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

## Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

# 4. Compare the cases

## CASE GALLERY

### Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.



**NORMAL**  
Expected learner request



**AMBIGUOUS**  
Two plausible meanings

NORMAL CASE: predict and name evidence. BOUNDARY CASE: predict and name evidence.



## MISSING SOURCE

Evidence unavailable



## ADVERSARIAL

Rules or trust challenged

MISSING INFORMATION: predict and name evidence. TRANSFER CASE: predict and name evidence.

Which case is most likely to reveal a hidden assumption? Why?

**Write or draw your reasoning and evidence.**

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### Change one thing in the pictured case

Change: LIB-v = 2 is unavailable.

Prediction: Report that the current time cannot be verified.

Reason: The unsupported old guess does not become current evidence when retrieval fails.

## 5. Words that unlock the idea

### VOCABULARY LAB

#### Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

1

**PARAMETERS**  
learned numeric patterns  
inside a model

2

## **RETRIEVAL**

search external stored  
sources

3

## **GROUNDING**

tie an answer to supplied  
evidence

4

## ABSTAIN

decline when evidence is insufficient

Word · My own example · Not this / boundary

**Word:** parameters

**My own example:** Values within a model adjusted during training; distinguish them from settings chosen by a developer.

**Not this / boundary:** \_\_\_\_\_

\_\_\_\_\_

**Word:** retrieval

**My own example:** Finding and returning existing information from a source or collection.

**Not this / boundary:** \_\_\_\_\_

\_\_\_\_\_

**Word:** grounding

**My own example:** Connecting an answer or action to specified evidence or observations.

**Not this / boundary:** \_\_\_\_\_

\_\_\_\_\_

**Word:** abstain

**My own example:** Decline to make a prediction when the available evidence does not meet the decision rule.

**Not this / boundary:** \_\_\_\_\_

\_\_\_\_\_

## Words used in the closer explanation

### Word · Meaning

**Word:** Corpus

**Meaning:** The collection of records available for a task

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**Word:** Index

**Meaning:** A searchable representation of that collection

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**Word:** Freshness

**Meaning:** Whether a record is current for the requested time

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## 6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

### WEEK 17 • WORKED EVIDENCE

- 1 Model memory may produce a plausible general explanation.
- 2 A current date or exact quotation requires retrieval.
- 3 Arithmetic should be calculated, not guessed from patterns.
- 4 When source or tool is unavailable, state the limit and abstain.

1. Model memory may produce a plausible general explanation.
2. A current date or exact quotation requires retrieval.
3. Arithmetic should be calculated, not guessed from patterns.

4. When source or tool is unavailable, state the limit and abstain.

Recalculate, retrace, or restate the reasoning in your own representation:

**Write or draw your reasoning and evidence.**

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CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

### A closer explanation

Index v1 contains notice N1 for room A. Corpus v2 supersedes it with N2 for room B, but the query still returns N1. What can be reported?

The stale index returns old evidence. It cannot establish current room approval; refresh against authorised corpus v2 or report unresolved freshness.

Compare the intermediate evidence with the final statement. Explain why each number or decision follows from the stated inputs. The conclusion belongs to this scenario and procedure; changing the inputs, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

### Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

### A pictorial worked case: Model Memory vs Retrieval

The library assistant remembers an old opening time, but a current notice can be retrieved.

#### The facts we will use

Approved notice LIB-v = 2 says Wednesday closes at 16: 00; the model previously produced 15: 00. The question asks the current Wednesday closing time.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

## Model Memory vs Retrieval

### Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



### 1. Old guess

A remembered-looking answer says 15: 00; no current notice is attached.



### 2. Find notice

Retrieve authorised LIB-v = 2, which explicitly says Wednesday closes at 16: 00.



### 3. Ground answer

Answer 16: 00 and attach LIB-v = 2. Its words, rather than the earlier guess, support this claim.

### CHANGE ONE THING

LIB-v = 2 is unavailable.

Explain the changed result using the matching lesson.

## Read the picture one step at a time

### Step · What happens and why

**Step:** 1. Old guess

**What happens and why:** A remembered-looking answer says 15: 00; no current notice is attached.

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**Step:** 2. Find notice

**What happens and why:** Retrieve authorised LIB-v = 2, which explicitly says Wednesday closes at 16: 00.

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### Step: 3. Ground answer

**What happens and why:** Answer 16: 00 and attach LIB-v = 2. Its words, rather than the earlier guess, support this claim.

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Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

### Why this result follows

Read the picture in order. Answer 16: 00 and attach LIB-v = 2. Its words, rather than the earlier guess, support this claim. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

### What this example does not establish

A retrieved document can itself be wrong or outdated; source approval, date and scope still need checking.

Read the labelled picture: Why can the assistant not certify 15: 00 when the notice is missing? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

Change one thing: LIB-v = 2 is unavailable. Predict the result before checking the explanation. What remains the same?

## 7. Find and repair a mistake

### ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak evidence.



### COMPARE • DIAGNOSE

Weak idea: A language model can inspect where every learned fact came from.  
Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

**Evidence record · Expected · Actual · Likely cause / next test**

**Evidence record:** normal case

**Expected:** works

**Actual:** \_\_\_\_\_

**Likely cause / next test:** \_\_\_\_\_

\_\_\_\_\_

**Evidence record:** boundary case

**Expected:** stated rule

**Actual:** \_\_\_\_\_

**Likely cause / next test:** \_\_\_\_\_

\_\_\_\_\_

**Evidence record:** missing input

**Expected:** safe response

**Actual:** \_\_\_\_\_

**Likely cause / next test:** \_\_\_\_\_

\_\_\_\_\_

**Evidence record:** fresh transfer

**Expected:** generalizes

**Actual:** \_\_\_\_\_

**Likely cause / next test:** \_\_\_\_\_

\_\_\_\_\_

Repair one cause. Why is this repair better than changing several things?

**Write or draw your reasoning and evidence.**

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

## A closer explanation

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer. For a project,

select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

**Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

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## 8. Practise together

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.



### SPECIFY THE SYSTEM

Define task, trusted context, constraints, schema, tools, checks, fallback, and stopping rule before trusting output.

SET task, audience, constraints, output\_schema LOAD approved\_context or retrieval\_result CALL model or allowed\_tool VALIDATE fields, claims, citations, and safety IF evidence missing: ask, retrieve, or abstain RETURN useful output + source trail + limitations

Trace one input. Which line changes the state or chooses a path?

**Write or draw your reasoning and evidence.**

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## A closer explanation

High retrieval similarity does not override permissions or make an obsolete source current.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself. The first may require a correction; the second requires an explicitly revised method and new evidence. Preserve unsuccessful cases so an improved result cannot hide the original problem.

**Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

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## Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Retrieval and indexing”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

### Plan · Do · Check

**Plan:** What is given? Which rule or procedure applies?

**Do:** Record each intermediate step.

**Check:** Does the result follow? What remains unknown?

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Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

## Check your reasoning

Use the worked case for Retrieval and indexing. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Why can the assistant not certify 15: 00 when the notice is missing? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

**My explanation using a pictured step and the given facts**  

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This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



**PREDICT → RUN → VERIFY → EXPLAIN**  
Route twelve questions to model, retrieval, calculator, or abstain; defend the evidence requirement for each.

Route twelve questions to model, retrieval, calculator, or abstain; defend the evidence requirement for each.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

Test / input: normal

Prediction: \_\_\_\_\_

Actual: \_\_\_\_\_

**Evidence / calculation:** \_\_\_\_\_

**Change or keep?:** \_\_\_\_\_

\_\_\_\_\_

**Test / input:** boundary

**Prediction:** \_\_\_\_\_

**Actual:** \_\_\_\_\_

**Evidence / calculation:** \_\_\_\_\_

**Change or keep?:** \_\_\_\_\_

\_\_\_\_\_

**Test / input:** missing

**Prediction:** \_\_\_\_\_

**Actual:** \_\_\_\_\_

**Evidence / calculation:** \_\_\_\_\_

**Change or keep?:** \_\_\_\_\_

\_\_\_\_\_

**Test / input:** fresh transfer

**Prediction:** \_\_\_\_\_

**Actual:** \_\_\_\_\_

**Evidence / calculation:** \_\_\_\_\_

**Change or keep?:** \_\_\_\_\_

\_\_\_\_\_

What did the hardest test teach you that the normal test did not?

**Write or draw your reasoning and evidence.**

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

**Plan → try → check**

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

# 10. Build your understanding

## MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the worked example.



CREATOR MASTERY

1 Define • 2 Trace • 3 Build • 4 Evaluate • 5 Verify • 6 Defend limits

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or evidence path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.
- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest evidence of independent understanding:

Write or draw your reasoning and evidence.

# 11. Connect your project

## CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



**VERIFIED STUDY BUDDY**

The Study Buddy retrieves course passages and never presents model memory as a citation.

**THIS WEEK’S CHECKPOINT** The Study Buddy retrieves course passages and never presents model memory as a citation.

**SYSTEM MAP** Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → human decision

What will you add, change, or verify in the project this week?

**Write or draw your reasoning and evidence.**

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What should remain under human control?

**Write or draw your reasoning and evidence.**

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**RESPONSIBLE DESIGN** Collect only necessary information. Show uncertainty and limits. Never confuse fluent output with verified truth.

## Evidence for your project

For your Verified Study Buddy, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

## 12. Show what you know

### INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



### INDEPENDENT CREATOR PROOF

Decide when to answer, retrieve, calculate, or abstain.

CHALLENGE Decide when to answer, retrieve, calculate, or abstain.

1. My fresh example, trace, calculation, or design:

**Write or draw your reasoning and evidence.**

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2. My evidence that it works (include a boundary or missing-data case):

**Write or draw your reasoning and evidence.**

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3. One limitation, fairness concern, or situation needing human review:

**Write or draw your reasoning and evidence.**

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SELF-CHECK ☐ I explained the mechanism ☐ I used evidence ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

## 13. Study a complete case

### Week 17 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

#### The situation

One assistant answers without search; another retrieves a handbook paragraph.

#### The question

Compare evidence trails.

#### Worked explanation

The first uses learned patterns without an inspected source. The second supplies a passage that can be checked. Retrieval still does not guarantee correct use.

#### Explain it in your own words

Which detail supports the answer, and why does it matter?

**Write or draw your reasoning and evidence.**

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## Trace the supplied case visually

### Retrieval creates an inspectable evidence trail

#### Evidence or stage · Value or result

**Evidence or stage:** No retrieval

**Value or result:** Answer from learned patterns

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**Evidence or stage:** Retrieval

**Value or result:** Inspect a handbook paragraph

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**Evidence or stage:** Verification

**Value or result:** Check the paragraph supports the answer

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One assistant answers without search; another retrieves a handbook paragraph.

Compare evidence trails.

The first uses learned patterns without an inspected source. The second supplies a passage that can be checked. Retrieval still does not guarantee correct use.

## 14. Try a new case independently

### Independent case: Apply a new case independently

Corpus has public P, teacher-only T and withdrawn W. A student search returns all three. Identify the filtering defect and whether hiding T after generation is sufficient.

#### My independent explanation, trace and limits

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Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

### Week 17 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A model recalls last year's opening hours; a retrieved current notice gives different hours. How should the answer use this evidence?

My answer, with a trace, calculation or supporting evidence

**Write or draw your reasoning and evidence.**

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### Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

**Write or draw your reasoning and evidence.**

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## Connect to your project

The Study Buddy retrieves course passages and never presents model memory as a citation.

The evidence I will keep

**Write or draw your reasoning and evidence.**

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Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

## Transfer practice from the original programme

A model states the library opens at 8 from learned patterns. A supplied current notice says 9. Explain the stronger answer and one remaining check.

**Explain the mechanism, evidence and one limit.**

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## Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

## 15. Your lesson route

Start with this question: The library assistant remembers an old opening time, but a current notice can be retrieved.

### By the end, I can · How I will show it

**By the end, I can:** Explain the mechanism and terminology of retrieval and indexing

**How I will show it:** Explain the deciding step, show a trace or test, and state one limit.

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**By the end, I can:** Apply the stated procedure to a new case with a traceable result

**How I will show it:** Explain the deciding step, show a trace or test, and state one limit.

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**By the end, I can:** Identify an assumption or limitation and design a revealing follow-up

**How I will show it:** Explain the deciding step, show a trace or test, and state one limit.

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Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

## 16. Finish and return

Close your notes. Explain this goal in your own words: Explain the mechanism and terminology of retrieval and indexing

### My deciding step, evidence and one remaining question

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If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.

## 17. Supported practice, repair and portfolio

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

The library assistant remembers an old opening time, but a current notice can be retrieved.

### Practice 1: explain the deciding evidence

Which labelled step matches this detail? Answer 16: 00 and attach LIB-v = 2. Its words, rather than the earlier guess, support this claim.

1. Old guess
2. Find notice
3. Ground answer

#### My selected conclusion, deciding fact and explanation

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Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Ground answer Answer 16: 00 and attach LIB-v = 2. Its words, rather than the earlier guess, support this claim. Read the picture in order. Answer 16: 00 and attach LIB-v = 2. Its words, rather than the earlier guess, support this claim. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

### Practice 2: explain the deciding evidence

Changed-case practice: LIB-v = 2 is unavailable. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. This one example guarantees correct results for every future input.
3. Report that the current time cannot be verified.

#### My selected conclusion, deciding fact and explanation

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Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Report that the current time cannot be verified. The unsupported old guess does not become current evidence when retrieval fails.

**Repair a missing building block**

Review Verify Numbers and Dates (ai-3/16). Goal: Explain the mechanism and terminology of claim support and citation evaluation. Short practice: Which part can be corrected immediately from existing evidence, and which needs follow-up?. Return to this week and explain what you changed.

Review Hallucination Failure Types (ai-3/14). Goal: Explain the mechanism and terminology of generation failures and verification. Short practice: Can we conclude there will definitely be no lunch from S alone?. Return to this week and explain what you changed.

Review Visual Composition (ai-3/11). Goal: Explain the mechanism and terminology of image generation and provenance. Short practice: Which checks pass after that single change, and what still needs repair?. Return to this week and explain what you changed.

**Keep your completed work**

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.